

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Bachelor of Computer Applications (BCA)****Semester-I University Examination MARCH -2018****CA 103 Operating Systems Concepts****Date: 26/03/2018, Monday Time:10.00 a.m. to 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. Attempt both sections in separate answer books.
2. The figure to the right indicates marks.
3. Make a suitable assumption when necessary.

SECTION – I**Q1 Do as directed.**

- [A] Define terms: Operating System, Process, System Call. [03]
- [B] List out objectives of operating system. [02]
- [C] Instruction Register store location of next instruction to be fetch. [01]
[True / False]
- [D] MBR stands for _____. [01]
- [E] Kernel is heart of operating system. [True / False] [01]
- [F] The main function of the command interpreter is to get and execute the next user-specified command. [True / False] [01]
- [G] First come first serve is pre-emptive scheduling algorithm.[True / False] [01]
- [H] The number of processes completed per unit time is known as Throughput. [True / False] [01]

Q2 Answer the followings. [Attempt any two]**[12]**

- [A] Explain services of operating system in detail.
- [B] Explain seven states process model with all transition.
- [C] List out and explain reasons of process termination in detail.
- [D] What is process control block? Explain in detail.

Q3 Answer the followings. [Attempt any two]**[12]**

- [A] Calculate turnaround time using First Come First Serve(FCFS) and Shortest Job First(SJF) scheduling algorithm.

Processes	Process Arrival Time	Service Time
P0	0	3
P1	2	6
P2	4	4
P3	6	8
P4	8	2

- [B] Explain five states process model with all transition.
- [C] Explain any two type of operating system in detail.
- [D] List and explain the functions of operating system.

SECTION – II

- Q4 Do as directed.**
- [A] Ver command is used to show version of DOS. [Yes / No] **[01]**
If no then justify your answer.
- [B] Write the output of `who | cut -d" " -f1` in Linux. **[02]**
- [C] Write output of `rename t1.txt t2.txt` in DOS. **[02]**
- [D] Write at least 3 use of `cat` command in Linux. **[03]**
- [E] List out at least three internal and external commands in DOS. **[03]**
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- Q5 Answer the followings. [Attempt any three]** **[12]**
- [A] Write a shell script to check whether enter number is odd or even.
- [B] What is the use of `cut` command in Linux? Explain it with example.
- [C] Write a shell script to calculate sum of 3 numbers.
- [D] What is the use of `chmod` command in Linux? Explain it with example.
- [E] What is the use of `date` command? Write a command to print date in format “23-03-2018” and “23-Mar-2018”.
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- Q6 Answer the followings. [Attempt Any Two]** **[12]**
- [A] What is use of `ls` command? Write five switches of `ls` with example.
- [B] Write a note on standard output , standard input and standard error.
- [C] Write a menu driven shell script to perform arithmetic operations between 2 numbers.
- [D] Write a shell script to print message “Good Morning” or “Good Afternoon” or “Good Evening” according to time.